
Certification Audit of the Cone-Scheme Decision Procedure

The $\forall$ PO-free fragment of $\mathcal{ALCI}_{\text{RCC5}}$, its full-logic boundary,
and its closest related work

Artifact reviewed: cone_scheme_attack.zip, supplied 2026-08-28
Report date: 2026-08-29
Technical author: OpenAI Codex, under the user's direction
Formal target: already-NNF concepts, abstract composition-table RCC5 semantics
Verdict: core proof chain survives cold attack; release certification is incomplete

Bottom line. The supplied Lean source contains a coherent finite greatest-fixed-point decision procedure and a fresh-occurrence soundness construction for the entire stated $\forall$ PO-free, mixed-quadrant fragment. No mathematical counterexample was found. This is not yet a release-grade certification: the source was not rebuilt in the present runtime, and the raw-syntax normalization, concrete-region adequacy bridge, pinned toolchain, capstone axiom transcript, and full regression gate are absent. The argument does not presently prove decidability of full $\mathcal{ALCI}_{\text{RCC5}}$; universal PO restrictions create a global all-pairs compatibility problem that the local cone signatures do not record.

Abstract

This report audits a claimed Lean decision procedure for the fragment of $\mathcal{ALCI}_{\text{RCC5}}$ in negation normal form containing no universal PO restriction. The procedure enumerates finite cone signatures, removes unsupported signatures by a monotone reductive operator, and unfolds any surviving signature into an infinite fresh-occurrence model. The audit traces the exact Lean declarations from syntax and semantics through completeness, ordered-disjoint geometry, the truth lemma, fixed-round correctness, and the final **Decidable** theorem. It independently reruns the supplied Python probes and separates their evidential value from the Lean theorem.

The core result appears mathematically sound for the semantics actually formalized: carrier-polymorphic frames whose five RCC5 atoms satisfy strong equality and the abstract composition table. The present packet does not formally identify these frames with a conventional nonempty-region semantics, and it accepts already-NNF syntax only. The report therefore assigns a qualified rather than final certification label.

An authorial generalization analysis identifies the exact failure at full logic. In the unfolding, every pair that is neither ordered nor disjoint is labelled PO. A formula $\forall \text{PO}.D$ consequently constrains arbitrary residual pairs, including cousins in unrelated generated branches. The current proof disposes of this case solely by the fragment hypothesis. A conditional PO-safe extension is straightforward, but full decidability requires new finite global interface machinery or a different automata argument. A staged Lean research plan and explicit go/no-go gates are given. Finally, the report compares the work with RCC constraint networks, composition-based spatial description logics, modal logics of topological relations, concrete-domain systems, type elimination, and unravelling. The finite GFP and fresh-witness ingredients are standard; no source was identified for the specific lower-spectrum cone signatures plus protected residual-PO unfolding or for this exact Lean result.

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1 Executive assessment

Within the explicitly formalized abstract semantics, the packet contains a genuine end-to-end decision theorem for every already-NNF concept satisfying POFree. The cold attack found no counterexample and no degenerate definition in the capstone route.

The appropriate public label is “substantively machine-formalized, independently source-audited, not yet reproducibly release-certified.” A clean Lean 4.32.0 rebuild and several semantic and packaging obligations remain.

The capstone declaration is:

```
def decidableSat_cone (C0 : Concept) (hpo : POFree C0) :
  Decidable (Satisfiable C0)
```

It has no oracle or semantic premise. The decision is a finite list computation; the model construction occurs only in its correctness proof. The strongest honest reading is therefore:

$$\forall C_0 \in \text{Concept}_{\text{NNF}}, \quad \text{POFree}(C_0) \implies \text{Decidable}(\text{Satisfiable}_{\text{absRCC5}}(C_0)).$$

The audit distinguishes four levels that should not be collapsed.

Level	Status	Meaning
Mathematical architecture	Passed cold attack	The finite GFP, source-model post-fixed point, occurrence geometry, and one-way truth lemma fit together.
Lean source chain	Present	All principal declarations through <code>decidableSat_cone</code> occur in the supplied source; static scans found no declaration-form axiom, <code>sorry</code> , <code>opaque</code> , or <code>unsafe</code> .
Independent build	Not performed	The present runtime has no Lean executable. The claimed Lean 4.32.0 build and axiom footprint were not reproduced.
External adequacy	Partial	The theorem targets abstract composition-table frames and already-NNF syntax. A raw-syntax theorem and a satisfaction-preserving bridge to ordinary set/region semantics are not packaged.

1.1 Answer to the full-logic question

The proof generalizes *structurally* but not yet *theoretically* to full $\mathcal{ALCI}_{RCC5}$. All finite-control, completeness, order, disjointness, existential, and non-PO universal machinery remains useful. One branch of the Lean truth lemma is missing: the case for $\forall \text{PO}.D$. Yet that single branch is not a local lemma. It asks that every residual PO pair in an infinite unfolding satisfy a two-sided type-compatibility condition. Pairs of separately born branches are not visible to either birth transition, so the current state cannot certify them.

The present proof therefore extends immediately only to instances equipped with an additional certificate that every residual PO pair is safe. It also extends to simple sufficient subfragments in which every PO-box body is globally true at all surviving types. It does not establish decidability for arbitrary $\forall \text{PO}$ formulas.

1.2 Answer to the novelty question

The responsible novelty claim is narrow. Type enumeration, monotone elimination, greatest fixed points, unravelling, and fresh witnesses are standard modal and description-logic methods. RCC5 composition and ordered/disjoint region representations are also established ideas. The apparently new contribution is their specific combination:

- a finite signature (T, S) consisting of a local support type and its possible strict-lower type spectrum;
- an inclusive-cone compatibility test that makes vertical and downward-DR propagation compositional;
- a monotone reductive elimination whose real-model signatures form a post-fixed point; and
- an infinite fresh-occurrence ordered-disjoint unfolding in which PO is residual and is sound precisely because universal PO boxes are absent.

No checked source was found that states the exact ∀PO-free theorem, uses this combination, or mechanizes it in Lean. This is evidence for novelty, not a universal priority guarantee; unpublished material or differently named equivalent machinery could still exist.

2 Exact formal scope

2.1 Syntax and semantics in the artifact

The **Concept** datatype is already in negation normal form. Atomic concept names and their negative literals are primitive; conjunction, disjunction, existential role restrictions, and universal role restrictions are constructors. The five roles are

$$RCC5 = \{EQ, DR, PO, PP, PPI\}.$$

The interpretation is carrier-polymorphic. **RCC5Interp** enforces strong identity for EQ, converse behavior, exhaustiveness/exclusivity, and the abstract RCC5 composition table. **Satisfiable C0** quantifies over a carrier, an interpretation, an RCC5 proof, and a designated domain point satisfying C_0 .

This is a legitimate and useful semantic target. It must nevertheless be named. A conventional spatial presentation begins with nonempty regions and defines PP by proper inclusion, DR by disjointness, PO by nonempty intersection without inclusion, and PPI by converse inclusion. The separate **RCC5NormalForm.lean** develops a canonical set representation for an ordered-disjoint structure, including inclusion, injectivity, and disjointness facts. It does not yet export the five concrete relations and prove a bidirectional satisfaction-preservation theorem connecting that representation to **POFreeLift.Satisfiable**.

2.2 What POFree means

POFree C0 recursively excludes every universal restriction with role PO. Existential PO demands remain allowed and are central to the mixed-quadrant theorem. The closure lemmas propagate the absence of a PO box to all relevant subformulas. Thus the result is not a PO-free logic: it is only ∀PO-free.

The distinction is the entire proof boundary. The soundness construction can always birth a fresh PO witness for $\exists PO.D$. What it cannot locally check is a box $\forall PO.D$ against every unrelated pair that later becomes residual PO.

2.3 What is not yet included

The packet does not currently provide:

- a raw concept syntax, an NNF transformation, or a theorem $\text{sat}(C) \leftrightarrow \text{sat}(\text{nnf}(C))$;
- a formal equivalence to a conventional nonempty-region/set semantics;
- a public Boolean `coneSatCoreB` and three-way wrapper that returns `outOfScope` on concepts containing a universal PO restriction;
- a formal closed expression for the advertised finite-state/complexity bound; or
- general TBoxes, nominals, number restrictions, role hierarchies beyond the fixed RCC5 table, or the full ∀PO language.

3 Why the cone scheme works

3.1 Finite control, infinite model

Let $\text{Cl}^+(C_0)$ be the finite positive closure and let $\text{Ty}(C_0)$ be the enumerated support types. A cone signature is

$$q = (T, S), \quad T \in \text{Ty}(C_0), \quad S \subseteq \text{Ty}(C_0),$$

where T is the local type and S is the spectrum of types that may occur strictly below the point in the generated proper-part order. The initial list $\text{sigStatic}(C_0)$ contains exactly the finitely checkable local candidates.

For a relation demand (r, D) of T , a target signature must carry D and satisfy `compatB`. Compatibility expresses the singleton composition consequences that must cross the birth edge. For example, proper-part edges transport relevant universal information into a whole cone, and a DR cut must remain disjoint from every descendant on both sides. PO births are protected from later being turned into order or disjointness.

Define the elimination operator

$$\Phi(X) = \{q \in X : \text{every demand of } q \text{ has a compatible target in } X\}.$$

It is reductive and monotone. Iteration from $\text{sigStatic}(C_0)$ reaches its greatest fixed point in at most the length of that list. Acceptance asks whether a surviving signature contains C_0 in its local type.

3.2 Completeness as a post-fixed-point argument

Given a genuine source model and a point x , form its exact signature: the local closure type realized at x , together with the set of realized types strictly below x . The artifact proves that exact signatures are static and that every real witness induces a compatible transition. Consequently the finite list of signatures realized in the model satisfies

$$X_M \subseteq \Phi(X_M).$$

By greatestness, X_M survives every elimination round. A source point satisfying C_0 therefore yields an accepting survivor. No finite submodel or quotient must be extracted; this avoids the cycle and shared-witness failures of earlier approaches.

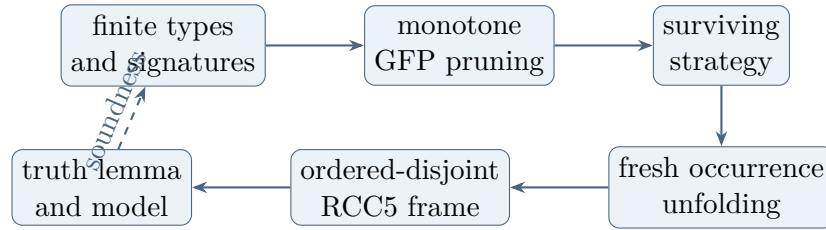
3.3 Soundness by fresh occurrences

Starting from a surviving signature, the construction recursively chooses a target for each demand. Every non-EQ demand creates a new occurrence even if its finite control signature has appeared before. The carrier is the subtype of generated finite demand words. Control may loop while domain points remain distinct.

The generated geometry is described by a strict order $<$ and a symmetric disjointness relation $\#$:

- vertical PP/PPI births generate $<$ and its transitive closure;
- DR births generate cuts whose descendant cones are placed in $\#$;
- order and disjointness are proved incompatible; and
- any unequal pair that is neither comparable nor disjoint receives residual PO.

The resulting five relations satisfy the abstract RCC5 frame axioms. An induction on concept structure then proves that every formula in an occurrence label is true there. All existential cases use the corresponding fresh child. The $\forall PP$, $\forall PPI$, $\forall DR$, and $\forall EQ$ cases use the cone and label propagation lemmas. The $\forall PO$ branch is impossible by **POFree**.



3.4 The decision theorem

The final proof avoids noncomputably choosing a stabilization round. It proves that iteration exactly $|\text{sigStatic}(C_0)|$ times is already fixed, establishes

$$\text{Satisfiable}(C_0) \iff \exists q \in \Phi^{|\text{sigStatic}(C_0)|}(\text{sigStatic}(C_0)), C_0 \in q.1,$$

and transfers the decidability of the finite right-hand side to satisfiability.

4 Lean theorem-chain audit

The source is a 44,375-line monolith containing several retired approaches before the cone scheme. The relevant capstone begins near line 41,861. Table 1 records the declarations that actually carry the result. Line numbers refer to the supplied `lean/POFreeLift.lean`.

Table 1: Principal Lean declarations in the cone-scheme chain.

Layer and lines	Declarations	Audit reading
Syntax/semantics, 378–420	Concept , Interp , sat , RCC5Interp , Satisfiable	Already-NNF syntax and abstract, carrier-polymorphic RCC5 composition-table semantics.

Layer and lines	Declarations	Audit reading
Fragment, 1143–1177	<code>POFree, pofreeB, pofreeB_iff</code>	Syntactic exclusion of every universal PO restriction and Boolean reflection.
Finite controls, 41861–41936	<code>Sig, supportB, sigOkB,</code>	Enumerated local types, lower spectra, and demand transitions.
Elimination, 41937–41976	<code>sigStatic, compatB, sigDemands</code>	Reductive monotone GFP and finite stabilization.
Model signatures, 42067–42269	<code>pruneSig, pruneSig_red,</code> <code>pruneSig_mono, coneScheme_gfp</code>	Realized signatures are static, transitions reflect witnesses, and the realized list is post-fixed.
Completeness, 42272	<code>dkey_sigOk, dkey_compat,</code> <code>modelSigs_survives,</code> <code>modelSigs_in_gfp</code> <code>coneScheme_complete</code>	Any source model yields an accepting survivor at every round. This theorem has no <code>POFree</code> premise.
Occurrence geometry, 42308–42903	<code>Occ, ostep, unfLt, unfDisj, gOD</code>	Fresh demand words, ranked strict order, symmetric downward disjointness, and generated carrier.
Strategy, 42506–42689	<code>pickTarget, pickTarget_some,</code> <code>pickTarget_closed, osig, Gen</code>	Computable target selection and closure of generated paths.
Propagation, 42918–43079	<code>allPP_gLt, allPPI_gLt,</code> <code>allEQ_local, allDR_gDisj,</code> <code>gchild_serves</code>	Every universal case except PO, plus all existential witness cases.
Interpretation, 43907–43914	<code>unfInterp, unfInterp_rcc5</code>	The generated occurrence structure is an abstract RCC5 interpretation.
Truth, 43937–44011	<code>unf_truth</code>	Membership-to-truth induction; the universal PO case is rejected using <code>POFree</code> .
Soundness, 44024	<code>coneScheme_sound</code>	A fixed survivor containing C_0 yields a model.
Characterization, 44038 and 44110	<code>coneScheme_correct,</code> <code>coneScheme_correct_at</code>	SAT is equivalent to acceptance at the named finite round.
Decision, 44092 and 44127	<code>gfpIter_bound_fixed,</code> <code>decidableSat_cone</code>	Fixed iteration bound and final Decidable result.

4.1 Static trust audit

A declaration-oriented source scan found no `sorry`, `sorryAx`, declaration-form `axiom`, `opaque`, or `unsafe`. Words such as “admits” in comments are not proof holes. The file contains 883 `#print axioms` commands, mostly for earlier machinery, but none targets the principal cone-scheme capstone declarations listed in Table 1. Thus the packet’s expected footprint of `propext`, `Quot.sound`, and `Classical.choice` remains a claim rather than an independently captured capstone transcript.

The packet includes neither `lean-toolchain` nor a Lake project. It states Lean 4.32.0 core with no Mathlib. Because no Lean executable was installed in the audit runtime, the source could not be independently elaborated. This is a reproducibility gap, not contrary mathematical evidence.

4.2 Separate normal-form file

The 231-line `RCC5NormalForm.lean` defines `RCC5Net`, an `OrderedDisjoint` structure, and a canonical representation η . It proves, among other facts, that canonical inclusion reproduces the order, that η is injective, and that canonical set disjointness reproduces the abstract disjointness relation. This strongly supports the adequacy route. It stops short of the complete bridge needed for a public concrete-region theorem: define all five set relations, prove residual PO equivalence, construct interpretations in both directions, and prove formula satisfaction is preserved.

5 Cold-attack evidence

5.1 Composition-table and geometry checks

The audit re-derived every one of the 25 RCC5 composition cells from nonempty finite-set semantics and compared the result with the artifact’s table; no mismatch was found. The occurrence construction was examined at its load-bearing invariants: vertical base preservation, increasing height on strict order, downward closure of DR cuts, incompatibility of order and disjointness, and protection of explicit PO births. No frame counterexample was found.

This evidence is subordinate to the Lean proof: finite sets validate the standard composition table, not every infinite model, and finite prefixes cannot establish an infinite truth lemma. Its purpose is to catch transcription and definition mistakes.

5.2 Rerun of wp134: pruning diagnostics

The supplied pruning probe exits 0. Its cap-free DR wall check reports 180 blocked type pairs and zero breaches. Its capped finite-state runs report:

Case	Initial	Survivors	Rounds	Expected	Outcome
U1	930	237	3	reject	reject
U2	3660	465	3	reject	reject
U3	930	273	4	reject	reject
S1	420	420	1	accept	accept
S2	86	28	3	accept	accept
S3	420	420	1	accept	accept
S4	420	288	2	accept	accept

The lower-spectrum enumeration is capped at $|S| \leq 1$. Acceptance is useful positive evidence, but rejection in the capped universe does not prove rejection in the complete signature space. The script is a hand transcription rather than execution of extracted Lean code.

5.3 Rerun of wp133: frame check and truth-column defect

The unfolding probe exits 1. For fresh occurrences it built 146 nodes at depth 2 and 144 nodes at depth 3, with zero reported frame violations and respectively four and five truth-column failures. Reuse-control variants also showed zero frame failures.

These truth failures do not refute `unf_truth`: the probe deliberately omits the signature/GFP compatibility assumptions and finite depth leaves existential demands unserved. `VERIFY.md` says not to treat the truth column as a check of the plan. The nonzero exit is nevertheless a packaging defect: a regression advertised under “How to check this” should either model its assumptions and pass or be explicitly classified as an expected negative/incomplete probe. The probe also does not directly test all ordered-disjoint axioms despite broad wording in its prose.

5.4 Adversarial conclusion

No concept was found for which the certified ∀PO-free checker has the wrong answer. The strongest residual concern is not a local cone-scheme flaw but semantic and release adequacy: whether the public phrase “ALCI-RCC5 satisfiability” is intended to mean exactly the already-NNF abstract semantics in the theorem.

6 Certification status against the prior ledger

The earlier plan named 24 obligations O00–O23. The new artifact discharges the mathematical spine largely through the intended construction, sometimes with a different interface. Table 2 is deliberately strict: “partial” means the result may be sufficient for the current theorem but not for the earlier release specification.

Table 2: Status of the 24-obligation certification ledger.

ID	Status	Evidence or remaining work
O00	Partial	Source supplied and archive intact; no pinned toolchain, clean-build transcript, theorem manifest, or supplied checksum manifest.
O01	Partial	NNF-only <code>Concept</code> , <code>P0Free</code> , and Boolean reflection exist; no raw syntax, normalization function, or semantic preservation theorem.
O02	Alternate	<code>supportB</code> soundness supplies the needed support-type facts; the exact proposed <code>Prop/Bool</code> interface and bitset representation are absent.
O03	Partial	Type, key, signature, and demand enumeration exist; no formal closed signature bound.
O04	Alternate	<code>compatB</code> is decoded by propagation lemmas and source-model compatibility, without a standalone <code>Prop/Bool</code> equivalence theorem.
O05	Bypassed	No abstract total <code>ConeScheme</code> structure; <code>pickTarget</code> and <code>osig</code> implement the needed behavior on generated paths.
O06	Done	Reductivity, monotonicity, finite stabilization, greatestness, and named round fixedness.
O07	Alternate	Strategy extraction is option-valued and closed on generated paths.
O08	Done by construction	Fresh demand words prevent domain-point reuse; a separate unique-parent API is not exported.
O09	Done	Integer-height strict order, transitivity, and irreflexivity.
O10	Done	Symmetric, irreflexive, downward-closed disjointness and incompatibility with order.
O11	Done	Explicit PO births remain neither ordered nor disjoint and become PO.

ID	Status	Evidence or remaining work
O12	Done	Ordered-disjoint frame and <code>unfInterp_rcc5</code> .
O13	Partial	Canonical set representation exists separately; no end-to-end concrete-region and satisfaction bridge.
O14	Done	Actual lower occurrence types lie in the declared cone spectrum.
O15	Done	One-way truth lemma <code>unf_truth</code> .
O16	Done in stated semantics	Surviving fixed signature yields an abstract RCC5 model.
O17	Done	Realized source signatures belong to the static set.
O18	Done	Real witnesses induce compatible source transitions.
O19	Done	Realized signatures survive every GFP round.
O20	Done in stated semantics	Fixed-round regular-strategy characterization.
O21	Partial	Genuine <code>Decidable</code> theorem; no public Boolean core, raw wrapper, or explicit out-of-scope result.
O22	Missing as specified	Two Python probes only; the intended Lean regression modules are absent and one probe exits nonzero.
O23	Missing	No pinned dual build, capstone axiom allowlist, code-generation test, dependency audit, or release manifest.

This table explains the verdict: the new artifact completes the research proof’s core, while the remaining tasks are principally interface, adequacy, complexity, regression, and release engineering. O13 is mathematically significant and should precede a claim about conventional spatial-region semantics.

7 Generalization to full $\mathcal{ALCI}_{RCC5}$

7.1 Exactly what survives

Most of the formal development is independent of `POFree`. In particular, the following already applies to arbitrary already-NNF concepts under the artifact’s abstract semantics:

- finite closure, types, signatures, transition checks, monotone pruning, and fixed-round computation;
- the source-model signature construction and its post-fixed-point proof;
- `coneScheme_complete`: satisfiability implies an accepting survivor at every iteration round;
- the fresh ordered-disjoint RCC5 frame;
- witnesses for all five existential roles; and
- truth propagation for $\forall EQ$, $\forall PP$, $\forall PPI$, and $\forall DR$.

An immediate certified corollary is easy to overlook: the present finite screen is a *one-sided full-logic refuter*. For any already-NNF input, if the fixed-round candidate set has no survivor carrying the root concept, then the concept is unsatisfiable. A survivor is conclusive only under `POFree`; with universal PO boxes it means “unknown.” A release wrapper should expose this useful three-way behavior rather than discard it.

7.2 The single Lean branch that becomes a global condition

For a local type T , write

$$B_{\text{PO}}(T) = \{D : \forall \text{PO}. D \in T\}.$$

Two endpoint types are safe for a PO edge when

$$\text{Safe}_{\text{PO}}(T, U)_{\text{quad}} \iff B_{\text{PO}}(T) \subseteq U \wedge B_{\text{PO}}(U) \subseteq T.$$

In the current ordered-disjoint interpretation,

$$u \text{ PO } v \iff u \neq v \wedge u \not\prec v \wedge v \not\prec u \wedge \neg(u \# v).$$

Thus every pair left unguarded by order or disjointness becomes PO. Such a pair can consist of cousins born by unrelated demands, far below the points whose transitions were locally checked. Full soundness would require

$$\forall u, v \in \text{Gen}, \quad u \text{ PO } v \implies \text{Safe}_{\text{PO}}(\text{lab}(u), \text{lab}(v)). \quad (\text{PO-Hintikka})$$

No current signature or birth rule establishes this all-pairs invariant. In `unf_truth`, the $\forall \text{PO}$ case is closed only by deriving a contradiction from `POFree`; `compatB po` deliberately carries no universal-PO propagation.

7.3 Two exact failure witnesses

The fragment premise cannot simply be deleted. The direct formula

$$C_{\text{dir}} = (\forall \text{PO}. A) \sqcap (\exists \text{PO}. \neg A)$$

is unsatisfiable, but the PO-blind finite checks admit a two-state post-fixed set: the root's PO existential is served by a type carrying $\neg A$, while no check enforces the root's PO box on that target.

More importantly, checking only explicit PO births is still insufficient. Put

$$Y = (\forall \text{PPI}. A) \sqcap (\forall \text{PO}. A), \quad C_{\text{joint}} = (\exists \text{PP}. Y) \sqcap (\exists \text{PO}. \neg A).$$

Suppose $x \text{ PPI } y$ and $x \text{ PO } z$ witness the two conjuncts. Then $y \text{ PPI } x$ and $x \text{ PO } z$, so the RCC5 composition table gives

$$\rho(y, z) \in \text{comp}(\text{PPI}, \text{PO}) = \{\text{PPI}, \text{PO}\}.$$

If $y \text{ PPI } z$, the PPI box in Y forces A at z ; if $y \text{ PO } z$, the PO box does. Both contradict $\neg A$ at z . Hence C_{joint} is unsatisfiable. Yet the current local scheme can independently choose a PP target for Y and a PO target for $\neg A$; all present transition tests pass. The fatal relation is between the two witnesses, not along either root birth.

These examples are included as a deterministic boundary probe in the accompanying package. They demonstrate that the obstruction is ternary and joint, not a missing one-edge body check.

7.4 Why an all-pairs patch loses completeness

A tempting repair is to require every pair of surviving signatures to be PO-safe. This is sound for the fixed residual geometry but too strong. Consider

$$C_{\text{sat}} = (\exists \text{PP}. (\forall \text{PO}. A)) \sqcap (\exists \text{PP}. \neg A).$$

It has a three-point chain model $xPPzPPy$. Both z and y are PP successors of x ; z carries $\neg A$; and y has no PO neighbor, so its PO box is vacuous. The current unfolding instead births the two PP witnesses as incomparable siblings and therefore labels their cross-pair PO. Rejecting that occurrence geometry would reject a satisfiable formula. Full logic needs the option to place independently born witnesses into a cross-order or DR relation when necessary.

There is also an algorithmic trap. A pruning clause that demands compatibility with *every* member of the current survivor set becomes easier when the set shrinks; it is antitone in precisely the argument for which the GFP proof needs monotonicity. This recreates the earlier project's nonmonotone-Q3 failure.

7.5 Expressivity warning: a universal modality

Because the five atoms partition all ordered pairs and EQ is identity, full logic defines a universal modality:

$$[U]D := \forall EQ.D \sqcap \forall DR.D \sqcap \forall PO.D \sqcap \forall PP.D \sqcap \forall PPI.D.$$

At any point, $[U]D$ forces D at every domain element. Universal PO therefore does more than add one symmetric role box: together with the other four boxes it permits global constraints and internalization of finite global assumptions. This does not by itself prove undecidability, but it is a concrete expressivity jump and another reason not to describe the full theorem as a small extension.

7.6 Precise present generalizations

The existing architecture supports three honest extensions now:

1. **Conditional PO-safe soundness.** Assume the generated unfolding satisfies (PO-Hintikka). The current truth proof then extends with one lemma `allPO_gResidual`; every other case is already available.
2. **Uniform-body subclasses.** Soundness holds when every body of every surviving universal PO restriction belongs to every surviving local type. This includes vacuous $\forall PO.\top$ and globally forced bodies. It is decidable but intentionally conservative.
3. **Full-logic refutation.** Since completeness is unrestricted, empty acceptance at the named round is a certified UNSAT answer for arbitrary inputs.

None is full decidability.

8 Research plan for full decidability and Lean certification

The plan below is a falsifiable research program, not a claim that the missing theorem already follows. Its central question is whether the necessary joint cross-relations admit a finite bounded interface. If that theorem fails, the cone architecture should not be stretched further; the next task is an undecidability reduction or a different automata model.

8.1 F0: formalize the boundary before redesign

Create `FullPO/Boundary.lean` with the three concepts above and prove:

- `not_sat_Cdir` and `not_sat_Cjoint` directly from the RCC5 table;

- explicit post-fixed survivor lists for the PO-blind scheme, showing why the current completeness screen cannot be used as a decision procedure;
- `sat_Csat` by a finite chain interpretation; and
- failure of PO-Hintikka for the current sibling unfolding of C_{sat} .

These become permanent regressions. Any proposed extension must reject the first two and accept the third.

8.2 F1: isolate a conditional full truth theorem

Define `POClosed X q0` exactly as (PO-Hintikka), with a Boolean endpoint-type version and a proof reflection theorem. Refactor `unf_truth` into:

```
theorem unf_truth_full
  (hstatic : ... ) (hfix : ... )
  (hpoClosed : POClosed X q0) :
  forall c u, c in lab u -> sat unfInterp u c
```

and derive the current result from `POFree`. This exposes the only missing semantic invariant without committing to a false finite approximation. It also yields a clean conditional theorem for user-supplied PO-safe strategies.

8.3 F2: replace independent targets by rooted RCC5 mosaics

A candidate finite control state should be a *mosaic* rather than one cone signature. For a root type T , include one designated representative for each existential demand, permit justified witness sharing, assign a support type and lower spectrum to every representative, and store a complete RCC5 atom matrix on the finite bag. A Boolean `mosaicOkB` must check:

- strong EQ, converse, and one atom per ordered pair;
- every RCC5 composition triangle wholly inside the bag;
- all universal bodies on every stored edge, including PO;
- inclusive-cone propagation for PP/PPI and downward DR closure;
- realization of every root demand by a designated member; and
- consistency and Boolean reflection of all local types.

The state space is finite for a fixed closure if the bag size is bounded. Prove the enumeration complete and give an explicit cardinality bound before implementing pruning.

8.4 F3: define positive continuation interfaces

Mosaics must be glued without later inventing unsafe cross-pairs. A continuation state should therefore expose an interface: the overlap bag, its full atom matrix, its labels, and the cone spectra visible across the boundary. Transitions are existential: “there exists a continuation mosaic agreeing on the interface.” This preserves the monotonicity needed for greatest-fixed-point elimination.

Pair states `Good(r, q, q')` are a useful first experiment, but RCC5 composition is ternary. The C_{joint} witness shows that pairwise birth checks can miss a triangle involving two children and their parent. Complete bounded bags are the safer primary design.

8.5 F4: prove source-model completeness for mosaics

From an arbitrary satisfying model, select witnesses for all demands at a point and record the exact RCC5 relations among the selected points. Prove that this yields a static mosaic and that every exposed continuation has a matching realized mosaic. The finite list of realized mosaics must be a post-fixed point of the new positive pruning operator. This stage is conceptually similar to the successful source-signature proof, but it depends on the chosen interface and witness-sharing policy.

8.6 F5: bounded-interface amalgamation – the go/no-go theorem

The load-bearing new theorem is:

For every closure, compatible surviving mosaics can be unfolded and amalgamated through interfaces of a computably bounded size into a total abstract RCC5 frame, with no unrecorded unsafe PO pair; conversely every model admits such bounded interfaces.

Proving only the forward direction gives soundness but may lose complete models; proving only the reverse direction gives a refuter but not a model. Both are required. The RCC5 triangle constraints may force interface growth, so the boundedness claim must be attacked before large-scale Lean implementation. Suitable methods to test are:

- a tree-decomposition theorem for realized witness networks;
- a finite forbidden-profile or Ramsey/well-quasi-order argument that folds equivalent boundary points;
- a guarded patchwork theorem for RCC5 mosaics; or
- an alternating/parity tree automaton whose states carry complete boundary bags.

Go/no-go gate: do not claim a full decision procedure and do not mechanize the final wrapper until a paper proof of bounded-interface amalgamation, in both directions, survives the three boundary regressions and an adversarial search for unbounded-width families.

If bounded interfaces provably fail, the failure should be exploited. Universal PO and the definable universal modality are natural ingredients for an undecidability reduction. A negative result would answer the full-logic question more honestly than an unsound finite-state approximation.

8.7 F6–F8: finish the theorem after the gate

Conditional on F5:

1. **F6, full soundness.** Unfold the accepting mosaic strategy, prove total RCC5 composition, prove PO-Hintikka for every generated cross-pair, and extend the five-role truth lemma.
2. **F7, finite decision.** Prove reductivity, monotonicity, finite stabilization, model completeness, fixed-round equivalence, a Boolean core, and a **Decidable** theorem for arbitrary already-NNF concepts.

3. **F8, external certification.** Add raw syntax/NNF preservation, the concrete-region adequacy bridge, a formal complexity bound, extracted execution tests, a pinned Lean toolchain, dual clean builds, exact axiom allowlists, and a checksum/theorem manifest.

The likely proof-engineering order is boundary tests $\rightarrow$ finite data and Boolean reflection $\rightarrow$ paper amalgamation theorem $\rightarrow$ source-model post-fixedness $\rightarrow$ unfolding geometry $\rightarrow$ truth lemma $\rightarrow$ decider. This keeps expensive formalization behind the mathematical gate.

9 Related work and novelty audit

9.1 Scope and method of the literature check

The audit compared the artifact with the project’s archived primary bibliography and the stable publication metadata listed below. The live scholarly-search backend was unavailable during this review, so this is not a claim to have searched every paper, thesis, or unpublished manuscript. The priority language is consequently conservative: “no exact precedent was identified in the direct literature checked.” A specialist review and a definition-level comparison with role-forming concrete-predicate logics remain publication gates.

9.2 Direct historical line

Cohn’s 1993 paper is an early source for treating qualitative spatial relations as modalities [2]. Wessel’s 2000–2003 reports study description logics with composition-based role constraints and introduce the direct $\mathcal{ALCI}_{\text{RCC}}$ problem family; the 2002/03 report records the RCC5 and RCC8 cases as unanswered [9, 10, 12]. This is the essential origin and priority context for the present project.

Lutz and Wolter’s modal logics of topological relations are the closest published direct comparator [15]. They analyze RCC-based modal languages and leave a prominent RCC5 region-structure case open. No $\forall$ PPO-free decision theorem, lower-spectrum cone signature, or fresh ordered-disjoint GFP construction was located there. However, the current artifact formalizes abstract composition-table frames. Its separate normal-form file does not yet prove that this semantics is exactly the region-structure semantics of Lutz and Wolter. The report therefore treats their work as the closest semantic precedent, not as a formally proved identity of logics.

9.3 RCC networks and relation algebra

The RCC relations and composition calculus originate in the region-connection tradition [1]. Jonsson and Drakengren classify tractability for RCC5 constraint languages [5]; Renz and Nebel analyze maximal tractable RCC fragments [6]; Dumentsch, Wang, and McCloskey develop a relation-algebraic account [11]; and Li and Ying examine models and the composition table [13]. These works supply the relation algebra, constraint-network setting, representations, and finite CSP boundary. They do not by themselves decide a modal description logic in which unary concept types, nested role quantification, and potentially infinite witness generation interact.

9.4 Concrete domains and role-forming predicates

Classical $\mathcal{ALC}(\mathcal{D})$ places concrete constraints on feature values; Baader and Hanschke provide the foundational integration scheme [3]. Later tableau and ontology results develop this line under

admissibility and patchwork conditions [16, 17]. Ordinary concrete-domain results must not be cited as direct solutions to a single-sorted complete RCC5 role partition.

There is an important close exception to any blanket claim that concrete-domain logics cannot form or quantify over relations. Haarslev, Lutz, and Moeller add a role-forming predicate operator that derives abstract roles from concrete predicates over feature chains [7]. Its feature-mediated, two-sorted semantics and safety conditions are visibly different from the artifact’s primitive, single-sorted, total five-atom role partition with strong equality and global composition closure. This review did not complete a definition-by-definition translation or separation proof. It therefore neither claims that their formalism subsumes the present logic nor that it cannot. Such an expressivity audit is mandatory before categorical “first result” language.

9.5 Standard proof machinery

Closure types, Hintikka labels, tableau expansion, inverse-role propagation, monotone elimination, finite control for infinite models, and fresh-witness unravelling are established modal and description-logic techniques. Representative sources include Schmidt-Schauss and Smolka’s concept-description calculus [4], Horrocks and Sattler on inverse/transitive roles [8], Donini and Massacci on EXPTIME tableaux [18], and Demri and de Nivelle on regular grammar logics with converse [14]. The current proof adapts these patterns; it did not invent type elimination or unravelling.

9.6 Event-structure analogy

After reversing the proper-part orientation, the ordered-disjoint normal form has the same relational skeleton as a prime event structure: strict order behaves like causality, downward-hereditary DR like conflict, and residual PO like concurrency. This structure is classical in concurrency theory [19, 20]. The analogy does not supply the RCC5-labelled modal truth lemma or the cone decision procedure. It does show that “order plus hereditary incompatibility plus residual concurrency” is not itself an unprecedented abstract geometry. The novelty lies in the RCC5 identification, type labelling, cone invariant, protected-PO argument, and decision proof built on it.

9.7 Component-by-component novelty verdict

Component	Assessment	Closest ancestry or caveat
RCC5 atoms and composition table	Inherited	Classical RCC and relation-algebra literature.
Closure/types and monotone GFP	Standard method	Modal/DL tableaux, filtration, and type-elimination traditions.
Fresh demand-word witnesses	Standard core, specialized use	Tableau/unravelling; the no-reuse discipline is adapted to avoid PP cycles.
Order, hereditary DR, residual PO	Close structural precedent	Prime event structures after reversing the order orientation.
Signature (T, S) with full strict-lower spectrum	Apparently new	No exact state representation identified in the direct literature.
Inclusive-cone and cross-DR compatibility	Apparently new combination	RCC5-specific use of singleton composition cells and all-descendant spectra.
Protected residual-PO lemma and exact no-$\forall\text{PO}$ boundary	Apparently new	No matching fragment theorem or proof boundary identified.
End-to-end fragment decidability theorem	Apparently new in checked corpus	Direct problem sources leave broader RCC5 cases open; role-forming predicate comparison still required.
Lean mechanization	Apparently new	No formal artifact for this exact logic/result was identified. Reproducible rebuild and adequacy bridge remain.

The strongest defensible publication sentence is:

The $\forall\text{PO}$ -free decidability theorem and its RCC5-specific cone machinery appear to be new relative to the direct literature located. The proof combines standard type elimination and unravelling with an event-structure-like ordered-conflict geometry. The priority claim remains subject to specialist review and a definition-level comparison with role-forming concrete-predicate logics.

9.8 Authorial and project provenance

The cone proof is best described as a project result produced through human-directed, multi-model theorem development and adversarial review. The present technical report and its full-logic analysis were authored by OpenAI Codex under the user’s direction. Earlier project rounds contributed the PP-kernel intuition, two-tier quotient attempts, and the adversarial discovery of the “PO gap” that fixed the present fragment boundary. The supplied design log records the immediate cone-scheme construction. This provenance supports credit and reproducibility; it is not a claim that any one model invented the standard ingredients ex nihilo.

10 Release-certification plan

The following sequence would turn the current source into a defensible archival artifact for the stated fragment. It is separate from the full-logic research program.

1. Add a `lean-toolchain` pin for Lean 4.32.0 and a minimal Lake project.
2. Extract the cone route into small modules, away from retired machinery; enable `set_option autoImplicit false` and document imports.
3. Add `#print axioms` for every capstone declaration in Table 1; retain exact output and compare it with a stated allowlist.
4. Perform two clean builds in independent environments and retain machine, compiler, elapsed-time, and peak-memory metadata.
5. Complete the ordered-disjoint-to-region bridge: five concrete relations, residual PO, representation in both directions, and satisfaction preservation.
6. Add raw syntax, inverse-role normalization, NNF transformation, and the semantic preservation theorem; state the public theorem over raw input.
7. Prove the signature cardinality and runtime upper bounds in Lean or clearly label the complexity estimate as external.
8. Export a Boolean `coneSatCoreB`; prove reflection; add a three-valued wrapper returning `sat`, `unsat`, or `outOfScope/unknown`.
9. Evaluate generated code on small formulas and audit that executable dependencies exclude the noncomputable model construction and retired routes.
10. Port the D1, D2, anchor, no-finite-model, mixed-control, protected-PO, DR, and role-aware demand regressions into Lean.
11. Repair `wp133` so its documented expected run exits 0, or rename it as an intentionally incomplete negative probe and remove it from the green gate.
12. Generate a theorem inventory, source manifest, SHA-256 checksums, build logs, probe transcripts, and a single `run_release_checks.sh` that fails closed.

The accompanying ZIP does not pretend these steps have already occurred. It preserves the reviewed source, adds this report and audit notes, includes deterministic probes and a checksum manifest, and makes each limitation explicit.

11 Conclusion

The cold attack did not break the cone-scheme theorem. The mathematical spine is substantial: finite lower-spectrum signatures, monotone GFP pruning, a post-fixed-point completeness proof, an infinite fresh-occurrence ordered-disjoint frame, a protected residual-PO construction, and a fixed-round decidability wrapper. For already-NNF $\forall$ PO-free concepts under abstract composition-table $RCC5$ semantics, the result should be treated as a serious machine-formalized decision procedure.

Two boundaries must remain in the headline. First, release certification and external adequacy are not complete until the source is rebuilt under a pin and linked formally to raw syntax and ordinary region semantics. Second, the result does not settle full $\mathcal{ALCI}_{RCC5}$. Universal PO boxes observe arbitrary residual cross-branch pairs. Exact counterexamples show that neither deleting the premise nor adding a local PO transition check works, while a global pairwise restriction loses complete models and threatens monotonicity.

The proof nevertheless advances the full problem. It supplies an unrestricted one-sided refuter, isolates the exact PO-Hintikka invariant, provides regressions that any extension must pass, and

points to rooted RCC5 mosaics with bounded continuation interfaces as the most disciplined next attack. The decisive unanswered theorem is bounded-interface amalgamation. Until it is proved or refuted, full decidability should remain open.

A Archive manifest

The reviewed original archive has size 515,404 bytes and SHA-256 46ff587cf8828dac6a05c9a92f0c6119d0d5146ce060b0f742441221eedfc9e1. `unzip -t` reported no compressed-data error. Table 3 records the eight source files before audit packaging.

Table 3: Original source manifest.

File	Bytes	Lines	SHA-256
ATTACK_PROMPT_3.md	5,847	111	8bec4f3bf2864986e7033524107378c1f7587285b2cfd6d5c4f287c295e53a
README.md	1,651	40	9c33131c50767f32cdc42292ee26807235881a8eb930fbbd34c1490ba28c6cd2
VERIFY.md	2,119	56	c1d75cf6ec8d160ef01b18390c6a738e96f2721b20805e3b7be4ed3eee66036e
design/DESIGN_266_286.md	46,784	829	a09e65754e00f23a8e21e05261f642e2519bd35a5b8dc73a9f4f5e76927c2c30
lean/POFreeLift.lean	2,078,018	44,375	1674f7e6ff77c527e59305b64e9ce1721ff5938abbd5a8d0f117c7e8b95c4413
lean/RCC5NormalForm.lean	9,572	231	438e8277891054c82050d2d1c83065100881b0630692b72c7c71a5ee80d127ad
probes/wp133_cone_scheme_unfolding.py	12,969	359	abd0cb5bc9a42a1cfc15bb2b1902a0847c94f9380788f849cd5858ca879e0109
probes/wp134_cone_scheme_prune.py	13,307	387	140d09153e0066ab87a67247018bf879ab70e0638266eceb8454cf12de4fc304

B Probe transcript summary

```
$ python3 probes/wp134_cone_scheme_prune.py
DR wall: 180 blocked type pairs; 0 breaches
U1 930 -> 237 reject      S1 420 -> 420 accept
U2 3660 -> 465 reject     S2 86 -> 28 accept
U3 930 -> 273 reject      S3 420 -> 420 accept
                          S4 420 -> 288 accept
exit 0

$ python3 probes/wp133_cone_scheme_unfolding.py
fresh depth 2: 146 built, 0 frame violations, 4 truth failures
fresh depth 3: 144 built, 0 frame violations, 5 truth failures
exit 1

$ python3 probes/wp_full_logic_boundary.py
DIRECT cap0: static=15, survive=15, rounds=1, ACCEPT
JOINT cap1: static=3450, survive=1410, rounds=2, ACCEPT
semantic contradiction cores and explicit post-fixed sets: verified
exit 0
```

The first two commands refer to the original probes; the third is an audit addition. Captured outputs and their audit interpretation are stored in the package under `audit/probe_logs/`.

C Suggested public theorem stack

After the fragment release tasks are completed, the public API should distinguish semantic layers and scope:

```

theorem sat_nnf_iff (C : RawConcept) :
  RawSat C <-> Satisfiable (nnf C)

theorem abstract_region_sat_iff (C : Concept) :
  Satisfiable C <-> RegionSatisfiable C

theorem coneSatCoreB_spec (C : Concept) (h : POFree C) :
  coneSatCoreB C = true <-> Satisfiable C

def coneCheck (C : RawConcept) : CheckResult
-- sat / unsat / outOfScope

def decidableRawSat_pofree (C : RawConcept)
  (h : POFree (nnf C)) : Decidable (RawSat C)

```

For the full language, the sound public result available before F5 is only a refuter:

```

theorem coneReject_full_sound (C : Concept) :
  noAcceptingSurvivor C = true -> Not (Satisfiable C)

```

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